

Measurement checks – HPT (Czech data)
Reliability, dimensionality, and presentism-contextualization contrast

HPT and Extremism project

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1 What this document does

This report checks whether our **Historical Perspective-Taking (HPT)** instrument behaves well **before** we run any hypothesis tests.

We do three things:

1. **Reliability:** Are the HPT subscales internally consistent? We report **Cronbach's alpha** (α) and **McDonald's omega** (ω) for **POP**, **ROA**, **CONT** (three items each; 1-4).
2. **Dimensionality (CFA/EFA):** Does the **factor structure** match prior research (e.g., **POP+CONT** vs **ROA**, or three correlated factors)?
3. **Presentism-contextualization contrast:** Do **POP** (presentist) and **CONT** (contextualization) show the expected contrast? > **Composite scores used** (for descriptives and later files): > - **POP_rev** = **5** - **POP_raw** (so higher = more contextualized). > - **HPT_CTX6** = **mean(POP_rev, CONT)** (default composite). > - **HPT_TOT9** = **mean(POP_rev, CONT, ROA)** (includes ROA; robustness).

2 Setup and data loading

```
options(width = 120)

# Data handling & plots
library(tidyverse)

# Psychometrics
library(psych)      # alpha, omega, polychoric, EFA helpers
library(lavaan)     # CFA
library(semTools)   # model comparisons & extras

# Tables
library(knitr)

# Make kableExtra use longtable/booktabs and avoid loading tabu
options(kableExtra.latex.load_packages = FALSE)
library(kableExtra)
source("scoring_helpers.R")

# Load the dataset created in 00_data-preparation
normalised_responses <- readRDS("student_responses.RDS")
stopifnot(exists("normalised_responses"))
dat <- normalised_responses

# POP reversed item-wise and subscale helper
POP_rev_items <- paste0("POP", 1:3)

# Reverse POP items (1-4)
```

```

dat <- dat %>%
  mutate(
    across(
      all_of(POP_rev_items),
      ~ 5 - suppressWarnings(as.numeric(.)),
      .names = "{.col}_rev"      # <<< THIS was the culprit
    )
  ) %>%
  mutate(
    HPT_POP_raw = scale_mean(., POP_rev_items, min_answered = 2),
    HPT_POP_rev = scale_mean(., paste0(POP_rev_items, "_rev"), min_answered = 2),
    HPT_CONT    = scale_mean(., paste0("CONT", 1:3), min_answered = 2),
    HPT_ROA     = scale_mean(., paste0("ROA", 1:3), min_answered = 2),
    HPT_CTX6    = rowMeans(cbind(HPT_POP_rev, HPT_CONT), na.rm = FALSE),
    HPT_TOT9    = rowMeans(cbind(HPT_POP_rev, HPT_CONT, HPT_ROA),
                          na.rm = FALSE)
  )

print_tbl <- function(df, caption, digits = 3, escape = TRUE) {
  kbl(df, booktabs = TRUE, longtable = TRUE, caption = caption, digits = digits, escape = escape) |>
  kable_styling(full_width = FALSE, latex_options = c("hold_position"))
}

```

We verify that **HPT items** exist. If something is missing, we stop with a clear message.

```

## -- check-columns -----
hpt_cols <- c(paste0("POP", 1:3), paste0("ROA", 1:3), paste0("CONT", 1:3))
need     <- hpt_cols
miss     <- setdiff(need, names(dat))
if (length(miss)) stop("Missing variables: ", paste(miss, collapse = ", "))

keep <- complete.cases(dat[, hpt_cols])

analysis_df <- dat[keep, hpt_cols] |>
  as_tibble()

nrow_all   <- nrow(dat)
nrow_keep  <- nrow(analysis_df)
cat("Rows in full data: ", nrow_all, "\n",
    "Rows kept (complete HPT items): ", nrow_keep, "\n", sep = "")

```

```

## Rows in full data: 293
## Rows kept (complete HPT items): 276

```

3 Step 1 – Descriptives and scale construction

Why: Simple summaries catch obvious data problems and help readers develop intuition.

```
hpt_items <- analysis_df %>% select(all_of(hpt_cols)) # 9 HPT items

# Subscales and composites with POP reversed for composites
hpt_scores <- hpt_items %>%
  mutate(
    POP_raw = rowMeans(select(., starts_with("POP")), na.rm = TRUE),
    ROA      = rowMeans(select(., starts_with("ROA")), na.rm = TRUE),
    CONT     = rowMeans(select(., starts_with("CONT")), na.rm = TRUE)
  ) %>%
  mutate(
    POP_rev = 5 - POP_raw,
    HPT_CTX6 = rowMeans(cbind(POP_rev, CONT), na.rm = TRUE),
    HPT_TOT9 = rowMeans(cbind(POP_rev, CONT, ROA), na.rm = TRUE)
  )

summary(select(hpt_scores, POP_raw, POP_rev, ROA, CONT, HPT_CTX6, HPT_TOT9))
```

	POP_raw	POP_rev	ROA	CONT	HPT_CTX6	HPT_TOT9
## Min.	:1.000	Min. :1.333	Min. :1.000	Min. :1.000	Min. :1.333	Min. :1.667
## 1st Qu.	:1.583	1st Qu.:2.333	1st Qu.:2.333	1st Qu.:2.333	1st Qu.:2.500	1st Qu.:2.444
## Median	:2.000	Median :3.000	Median :3.000	Median :2.667	Median :2.833	Median :2.889
## Mean	:2.029	Mean :2.971	Mean :2.793	Mean :2.713	Mean :2.842	Mean :2.826
## 3rd Qu.	:2.667	3rd Qu.:3.417	3rd Qu.:3.333	3rd Qu.:3.333	3rd Qu.:3.333	3rd Qu.:3.222
## Max.	:3.667	Max. :4.000	Max. :4.000	Max. :4.000	Max. :4.000	Max. :3.889

```
cor(select(hpt_scores, POP_raw, ROA, CONT, HPT_CTX6, HPT_TOT9), use = "pairwise.complete.obs")
```

	POP_raw	ROA	CONT	HPT_CTX6	HPT_TOT9
## POP_raw	1.0000000	-0.1530998	-0.2763074	-0.7695631	-0.6452167
## ROA	-0.1530998	1.0000000	0.3694237	0.3351715	0.7105332
## CONT	-0.2763074	0.3694237	1.0000000	0.8263467	0.7871798
## HPT_CTX6	-0.7695631	0.3351715	0.8263467	1.0000000	0.9011123
## HPT_TOT9	-0.6452167	0.7105332	0.7871798	0.9011123	1.0000000

4 Step 2 – Reliability: α and ω for POP-ROA-CONT

```
alpha_poly <- function(x) {
  pc <- psych::polychoric(x)$rho
  psych::alpha(pc, n.obs = nrow(x))
}
```

```

}
omega_poly <- function(x) {
  pc <- psych::polychoric(x)$rho
  psych::omega(pc, n.obs = nrow(x), nfactors = 1, plot = FALSE)
}

subsets <- list(
  POP = hpt_items %>% select(starts_with("POP")),
  ROA = hpt_items %>% select(starts_with("ROA")),
  CONT = hpt_items %>% select(starts_with("CONT"))
)

rel_table <- purrr::imap_dfr(subsets, function(df, nm){
  a_raw <- psych::alpha(df)
  a_poly <- alpha_poly(df)
  om <- omega_poly(df)
  tibble(
    scale = nm,
    k_items = ncol(df),
    alpha_raw = unname(a_raw$total$raw_alpha),
    alpha_poly = unname(a_poly$total$raw_alpha),
    omega_total = unname(om$omega.tot),
    omega_hier = unname(om$omega.h)
  )
})

## Loading required namespace: GPArotation
## Omega_h for 1 factor is not meaningful, just omega_t
## Omega_h for 1 factor is not meaningful, just omega_t
## Omega_h for 1 factor is not meaningful, just omega_t
print_tbl(rel_table, digits = 3, caption = "Reliability of HPT subscales (alpha and omega).")

```

Table 1: Reliability of HPT subscales (alpha and omega).

scale	k_items	alpha_raw	alpha_poly	omega_total
POP	3	0.466	0.525	0.546
ROA	3	0.502	0.525	0.558
CONT	3	0.636	0.689	0.693

5 Step 3 – Dimensionality (CFA/EFA)

```
hpt_ord <- hpt_items  # treat items as ordered

m1_2factor <- '
F1 =~ POP1 + POP2 + POP3 + CONT1 + CONT2 + CONT3
F2 =~ ROA1 + ROA2 + ROA3
F1 ~~ F2
'

m2_3factor <- '
POP  =~ POP1 + POP2 + POP3
CONT =~ CONT1 + CONT2 + CONT3
ROA  =~ ROA1 + ROA2 + ROA3
POP  ~~ CONT + ROA
CONT ~~ ROA
'

m3_1factor <- '
G =~ POP1 + POP2 + POP3 + ROA1 + ROA2 + ROA3 + CONT1 + CONT2 + CONT3
'

fit_2 <- cfa(m1_2factor, data = hpt_ord, ordered = hpt_cols, estimator = "WLSMV")
fit_3 <- cfa(m2_3factor, data = hpt_ord, ordered = hpt_cols, estimator = "WLSMV")
fit_1 <- cfa(m3_1factor, data = hpt_ord, ordered = hpt_cols, estimator = "WLSMV")

# Compare fits
semTools::compareFit(fit_2, fit_3, fit_1)

## The following lavaan models were compared:
##      fit_3
##      fit_2
##      fit_1
## To view results, assign the compareFit() output to an object and use the summary() method; see the class?FitDiff help page.

report_fit <- function(fit) {
  list(
    indices = fitMeasures(fit, c(
      "chisq.scaled", "df.scaled", "pvalue.scaled", "cfi.scaled",
      "tli.scaled", "rmsea.scaled", "rmsea.ci.lower.scaled",
      "rmsea.ci.upper.scaled", "rmsea.pvalue.scaled", "srmr"
    )),
    loadings = standardizedSolution(fit) %>% as_tibble() %>% filter(op == "=~")
  )
}
```

```

cfa_summary <- list(
  `2-factor (POP+CONT vs ROA)` = report_fit(fit_2),
  `3-factor (POP/CONT/ROA)`    = report_fit(fit_3),
  `1-factor (general)`         = report_fit(fit_1)
)

purrr::iwalk(cfa_summary, function(x, nm){
  cat("\n###", nm, "\n")
  print(x$indices)
  print(kable(x$loadings, digits = 3))
})

```

```

##
## ### 2-factor (POP+CONT vs ROA)
##          chisq.scaled      df.scaled      pvalue.scaled      cfi.scaled      tli.scaled
##          61.297           26.000           0.000           0.913           0.880
##          rmsea.scaled rmsea.ci.lower.scaled rmsea.ci.upper.scaled rmsea.pvalue.scaled      srmr
##          0.070           0.048           0.093           0.069           0.066
##
##
## |lhs|op|rhs| |est.std| |se| |z| pvalue| ci.lower| ci.upper|
## |----|:--|:-----|-----:|-----:|-----:|-----:|-----:|-----:|
## |F1| |=~|POP1| | -0.489| 0.066| -7.385| 0.000| -0.619| -0.359|
## |F1| |=~|POP2| | -0.191| 0.068| -2.822| 0.005| -0.324| -0.058|
## |F1| |=~|POP3| | -0.358| 0.061| -5.907| 0.000| -0.477| -0.239|
## |F1| |=~|CONT1| | 0.690| 0.055| 12.630| 0.000| 0.583| 0.797|
## |F1| |=~|CONT2| | 0.595| 0.058| 10.258| 0.000| 0.481| 0.709|
## |F1| |=~|CONT3| | 0.618| 0.054| 11.416| 0.000| 0.512| 0.724|
## |F2| |=~|ROA1| | 0.604| 0.072| 8.410| 0.000| 0.463| 0.745|
## |F2| |=~|ROA2| | 0.293| 0.075| 3.916| 0.000| 0.146| 0.439|
## |F2| |=~|ROA3| | 0.676| 0.075| 9.016| 0.000| 0.529| 0.822|
##
## ### 3-factor (POP/CONT/ROA)
##          chisq.scaled      df.scaled      pvalue.scaled      cfi.scaled      tli.scaled
##          34.258           24.000           0.080           0.975           0.962
##          rmsea.scaled rmsea.ci.lower.scaled rmsea.ci.upper.scaled rmsea.pvalue.scaled      srmr
##          0.039           0.000           0.067           0.702           0.048
##
##
## |lhs|op|rhs| |est.std| |se| |z| pvalue| ci.lower| ci.upper|
## |----|:--|:-----|-----:|-----:|-----:|-----:|-----:|-----:|
## |POP| |=~|POP1| | 0.735| 0.099| 7.445| 0| 0.542| 0.929|

```

```
## |POP | =~ |POP2 | | 0.295| 0.076| 3.865|      0| 0.145| 0.444|
## |POP | =~ |POP3 | | 0.501| 0.068| 7.414|      0| 0.368| 0.633|
## |CONT | =~ |CONT1 | | 0.716| 0.055| 13.084|     0| 0.608| 0.823|
## |CONT | =~ |CONT2 | | 0.609| 0.059| 10.395|     0| 0.494| 0.724|
## |CONT | =~ |CONT3 | | 0.637| 0.054| 11.867|     0| 0.532| 0.743|
## |ROA | =~ |ROA1 | | 0.606| 0.071| 8.524|      0| 0.467| 0.745|
## |ROA | =~ |ROA2 | | 0.296| 0.074| 3.972|      0| 0.150| 0.442|
## |ROA | =~ |ROA3 | | 0.672| 0.074| 9.070|      0| 0.527| 0.817|
##
## ### 1-factor (general)
##          chisq.scaled      df.scaled      pvalue.scaled      cfi.scaled      tli.scaled
##          81.538          27.000          0.000          0.866          0.821
##          rmsea.scaled rmsea.ci.lower.scaled rmsea.ci.upper.scaled rmsea.pvalue.scaled      srmr
##          0.086          0.065          0.107          0.003          0.075
##
##
## |lhs |op |rhs | | est.std|    se|      z| pvalue| ci.lower| ci.upper|
## |:---|:-|:---|:-----|-----|-----|:-----|:-----|:-----|
## |G | =~ |POP1 | | 0.479| 0.066| 7.262| 0.000| 0.349| 0.608|
## |G | =~ |POP2 | | 0.177| 0.067| 2.649| 0.008| 0.046| 0.308|
## |G | =~ |POP3 | | 0.344| 0.061| 5.679| 0.000| 0.225| 0.462|
## |G | =~ |ROA1 | | -0.467| 0.061| -7.595| 0.000| -0.587| -0.346|
## |G | =~ |ROA2 | | -0.227| 0.067| -3.397| 0.001| -0.358| -0.096|
## |G | =~ |ROA3 | | -0.515| 0.059| -8.782| 0.000| -0.630| -0.400|
## |G | =~ |CONT1 | | -0.672| 0.053| -12.761| 0.000| -0.775| -0.569|
## |G | =~ |CONT2 | | -0.583| 0.057| -10.319| 0.000| -0.694| -0.472|
## |G | =~ |CONT3 | | -0.596| 0.052| -11.471| 0.000| -0.697| -0.494|
```

5.0.1 Optional: EFA (polychoric)

```
pc <- psych::polychoric(hpt_ord)$rho
efa2 <- psych::fa(pc, nfactors = 2, fm = "pa", rotate = "oblimin")
efa3 <- psych::fa(pc, nfactors = 3, fm = "pa", rotate = "oblimin")

cat("\nEFA (2 factors):\n")

##
## EFA (2 factors):
print(efa2$loadings, cutoff = 0.25)

##
## Loadings:
```

```
##          PA2    PA1
## POP1  -0.288  0.371
## POP2           0.448
## POP3           0.635
## ROA1   0.563
## ROA2   0.356
## ROA3   0.579
## CONT1  0.631
## CONT2  0.527
## CONT3  0.521
##
##          PA2    PA1
## SS loadings    1.811 0.833
## Proportion Var 0.201 0.093
## Cumulative Var 0.201 0.294
```

```
cat("\nEFA (3 factors):\n")
```

```
##
## EFA (3 factors):
```

```
print(efa3$loadings, cutoff = 0.25)
```

```
##
## Loadings:
##          PA1    PA3    PA2
## POP1           0.397
## POP2           0.382
## POP3           0.763
## ROA1           0.577
## ROA2           0.456
## ROA3           0.585
## CONT1  0.507
## CONT2  0.396
## CONT3  0.816
##
##          PA1    PA3    PA2
## SS loadings    1.109 1.035 0.912
## Proportion Var 0.123 0.115 0.101
## Cumulative Var 0.123 0.238 0.340
```

6 Step 4 – Presentism-contextualization contrast (POP vs CONT)

```
contrast_tbl <- hpt_scores %>%
  summarise(
    mean_POP = mean(POP_raw, na.rm = TRUE), sd_POP = sd(POP_raw, na.rm = TRUE),
    mean_CONT = mean(CONT, na.rm = TRUE), sd_CONT = sd(CONT, na.rm = TRUE),
    r_POP_CONT = cor(POP_raw, CONT, use = "pairwise.complete.obs")
  )

print_tbl(contrast_tbl, digits = 3, caption = "POP (raw) vs CONT: means, SDs, and correlation.")
```

Table 2: POP (raw) vs CONT: means, SDs, and correlation.

mean_POP	sd_POP	mean_CONT	sd_CONT	r_POP_CONT
2.029	0.646	2.713	0.732	-0.276

```
t_test <- t.test(hpt_scores$POP_raw, hpt_scores$CONT, paired = TRUE)
t_test
```

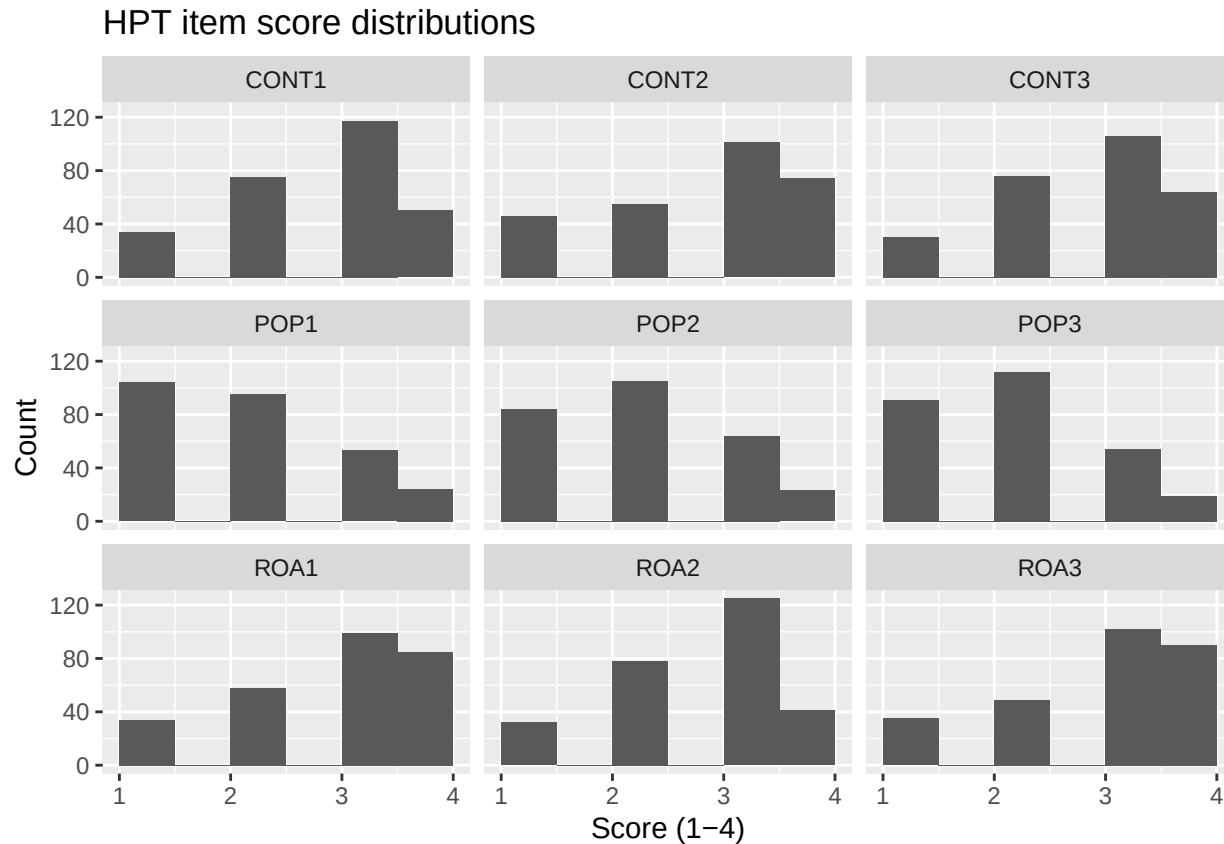
```
##
## Paired t-test
##
## data: hpt_scores$POP_raw and hpt_scores$CONT
## t = -10.306, df = 275, p-value < 2.2e-16
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -0.8141513 -0.5529985
## sample estimates:
## mean difference
## -0.6835749
```

7 Step 5 – Distribution checks

```
# Item distributions
long_items <- hpt_items %>%
  pivot_longer(cols = everything(), names_to = "item", values_to = "score")

ggplot(long_items, aes(score)) +
  geom_histogram(binwidth = 0.5, boundary = 0, closed = "left") +
```

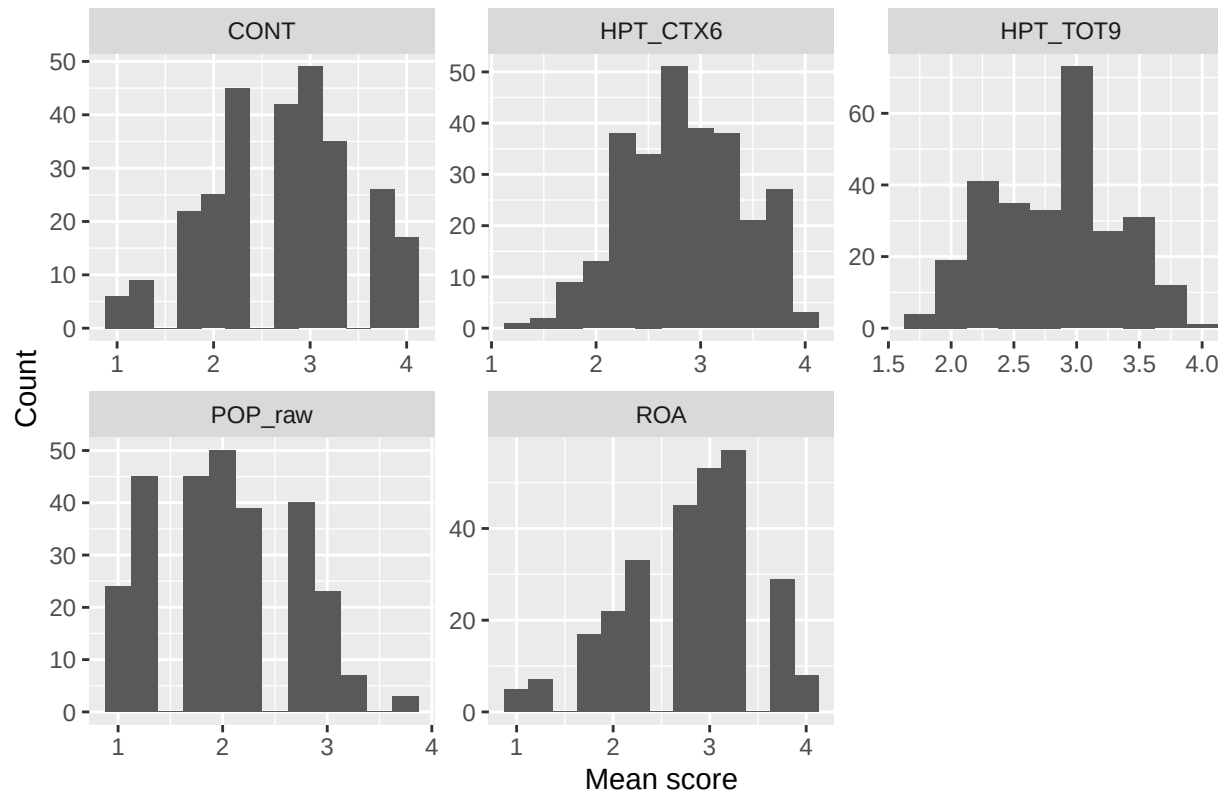
```
facet_wrap(~ item, ncol = 3) +
labs(title = "HPT item score distributions", x = "Score (1-4)", y = "Count")
```



```
# Scale/composite distributions
long_scales <- hpt_scores %>%
  select(POP_raw, ROA, CONT, HPT_CTX6, HPT_TOT9) %>%
  pivot_longer(everything(), names_to = "scale", values_to = "score")

ggplot(long_scales, aes(x = score)) +
  geom_histogram(binwidth = 0.25) +
  facet_wrap(~ scale, scales = "free") +
  labs(title = "Subscales and composites", x = "Mean score", y = "Count")
```

Subscales and composites



8 Step 6 – Knowledge mini-test (KN1-KN6)

```
kn_cols <- paste0("KN", 1:6)
has_kn <- all(kn_cols %in% names(dat))

if (!has_kn) {
  cat("\n**Knowledge section skipped:** KN1-KN6 not found in data.\n")
} else {
  kn_items <- dat[keep, kn_cols] # align to analysis_df rows via 'keep'
  # Basic sanity: coerce to numeric 0/1
  kn_items <- kn_items %>% mutate(across(everything(), ~ as.numeric(.)))

  # Total score, difficulty (p), discrimination (point-biserial)
  kn_total <- rowSums(kn_items, na.rm = TRUE)
```

```

item_stats <- tibble(
  item = kn_cols,
  difficulty_p = sapply(kn_items, function(x) mean(x, na.rm = TRUE)),
  discr_pb = sapply(kn_items, function(x) cor(x, kn_total - x, use = "pairwise.complete.obs"))
)

# KR-20 (alpha on dichotomous items)
kn_alpha <- psych::alpha(kn_items)

print_tbl(item_stats, digits = 3, caption = "KN items: difficulty (p) and point-biserial discrimination.")

print_tbl(tibble(
  k_items = ncol(kn_items),
  total_mean = mean(kn_total, na.rm = TRUE),
  total_sd = sd(kn_total, na.rm = TRUE),
  alpha_KR20 = unname(kn_alpha$total$raw_alpha)
), digits = 3, caption = "KN total: summary and KR-20 (alpha for dichotomous items).")
}

```

Table 3: KN total: summary and KR-20 (alpha for dichotomous items).

k_items	total_mean	total_sd	alpha_KR20
6	3.029	1.622	0.544

9 Step 8 – Ideology batteries (FR-LF mini, KSA-3) and Social Desirability (SDR-5)

```

# Helper: reliability table for Likert batteries (polychoric + omega total)
alpha_poly_likert <- function(x) {
  pc <- psych::polychoric(x)$rho
  psych::alpha(pc, n.obs = nrow(x))
}

omega_total_poly_likert <- function(x) {
  pc <- psych::polychoric(x)$rho
  if (!all(eigen(pc, symmetric = TRUE)$values > 1e-6)) pc <- psych::cor.smooth(pc)
  suppressWarnings(psych::omega(pc, n.obs = nrow(x), nfactors = 1, plot = FALSE)$omega.tot)
}

```

9.1 FR-LF mini (RD1-RD3, NS1-NS3)

```
fr_cols <- c(paste0("RD", 1:3), paste0("NS", 1:3))
has_fr <- all(fr_cols %in% names(dat))

if (!has_fr) {
  cat("\n**FR-LF mini section skipped:** RD1-3 and/or NS1-3 not found.\n")
} else {
  fr_df <- dat[keep, fr_cols] %>% as_tibble()
  RD <- fr_df %>% select(starts_with("RD"))
  NS <- fr_df %>% select(starts_with("NS"))

  fr_rel <- bind_rows(
    {
      a <- psych::alpha(RD); ap <- alpha_poly_likert(RD); wt <- omega_total_poly_likert(RD)
      tibble(scale = "FR-LF: RD", k_items = ncol(RD),
              alpha_raw = a$total$raw_alpha, alpha_poly = ap$total$raw_alpha, omega_total = wt)
    },
    {
      a <- psych::alpha(NS); ap <- alpha_poly_likert(NS); wt <- omega_total_poly_likert(NS)
      tibble(scale = "FR-LF: NS", k_items = ncol(NS),
              alpha_raw = a$total$raw_alpha, alpha_poly = ap$total$raw_alpha, omega_total = wt)
    },
    {
      a <- psych::alpha(fr_df); ap <- alpha_poly_likert(fr_df); wt <- omega_total_poly_likert(fr_df)
      tibble(scale = "FR-LF: total (RD+NS)", k_items = ncol(fr_df),
              alpha_raw = a$total$raw_alpha, alpha_poly = ap$total$raw_alpha, omega_total = wt)
    }
  )

  print_tbl(fr_rel, digits = 3, caption = "FR-LF mini reliability (alpha, polychoric alpha, omega total).")

  # Optional CFA: 2 correlated factors (RD, NS), ordered WLSMV
  fr_model <- '
  RD =~ RD1 + RD2 + RD3
  NS =~ NS1 + NS2 + NS3
  RD ~~ NS
  '

  fr_fit <- try(lavaan::cfa(fr_model, data = fr_df, ordered = colnames(fr_df), estimator = "WLSMV"), silent = TRUE)
  if (!inherits(fr_fit, "try-error")) {
    print(fitMeasures(fr_fit, c(
      "chisq.scaled", "df.scaled", "pvalue.scaled", "cfi.scaled",

```

```

      "tli.scaled", "rmsea.scaled", "rmsea.ci.lower.scaled",
      "rmsea.ci.upper.scaled", "rmsea.pvalue.scaled", "srmr"
    )))
  } else {
    cat("\nFR-LF CFA skipped (model failed to converge).\n")
  }
}

```

```

## Omega_h for 1 factor is not meaningful, just omega_t
## Omega_h for 1 factor is not meaningful, just omega_t
## Omega_h for 1 factor is not meaningful, just omega_t

```

##	chisq.scaled	df.scaled	pvalue.scaled	cfi.scaled	tli.scaled
##	29.384	8.000	0.000	0.939	0.886
##	rmsea.scaled	rmsea.ci.lower.scaled	rmsea.ci.upper.scaled	rmsea.pvalue.scaled	srmr
##	0.101	0.063	0.141	0.015	0.055

9.2 KSA-3 (A1-A3, U1-U3, K1-K3)

```

ksa_cols <- c(paste0("A",1:3), paste0("U",1:3), paste0("K",1:3))
has_ksa <- all(ksa_cols %in% names(dat))

if (!has_ksa) {
  cat("\n**KSA-3 section skipped:** A1-A3, U1-U3, and/or K1-K3 not found.\n")
} else {
  ksa_df <- dat[keep, ksa_cols] %>% as_tibble()
  A <- ksa_df %>% select(starts_with("A"))
  U <- ksa_df %>% select(starts_with("U"))
  K <- ksa_df %>% select(starts_with("K"))

  ksa_rel <- bind_rows(
    {
      a <- psych::alpha(A); ap <- alpha_poly_likert(A); wt <- omega_total_poly_likert(A)
      tibble(scale = "KSA-3: Aggression (A)", k_items = 3,
        alpha_raw = a$total$raw_alpha, alpha_poly = ap$total$raw_alpha, omega_total = wt)
    },
    {
      a <- psych::alpha(U); ap <- alpha_poly_likert(U); wt <- omega_total_poly_likert(U)
      tibble(scale = "KSA-3: Submission (U)", k_items = 3,
        alpha_raw = a$total$raw_alpha, alpha_poly = ap$total$raw_alpha, omega_total = wt)
    },
    {

```

```

a <- psych::alpha(K); ap <- alpha_poly_likert(K); wt <- omega_total_poly_likert(K)
tibble(scale = "KSA-3: Conventionalism (K)", k_items = 3,
        alpha_raw = a$total$raw_alpha, alpha_poly = ap$total$raw_alpha, omega_total = wt)
},
{
a <- psych::alpha(ksa_df); ap <- alpha_poly_likert(ksa_df); wt <- omega_total_poly_likert(ksa_df)
tibble(scale = "KSA-3: total", k_items = 9,
        alpha_raw = a$total$raw_alpha, alpha_poly = ap$total$raw_alpha, omega_total = wt)
}
)

print_tbl(ksa_rel, digits = 3, caption = "KSA-3 reliability (alpha, polychoric alpha, omega total).")

# Optional CFA: 3 correlated factors (A, U, K)
ksa_model <- '
A =~ A1 + A2 + A3
U =~ U1 + U2 + U3
K =~ K1 + K2 + K3
A ~~ U + K
U ~~ K
'

ksa_fit <- try(lavaan::cfa(ksa_model, data = ksa_df, ordered = colnames(ksa_df), estimator = "WLSMV"), silent = TRUE)
if (!inherits(ksa_fit, "try-error")) {
  print(fitMeasures(ksa_fit, c(
    "chisq.scaled", "df.scaled", "pvalue.scaled", "cfi.scaled",
    "tli.scaled", "rmsea.scaled", "rmsea.ci.lower.scaled",
    "rmsea.ci.upper.scaled", "rmsea.pvalue.scaled", "srmr"
  )))
} else {
  cat("\nKSA-3 CFA skipped (model failed to converge).\n")
}
}

```

```

## Omega_h for 1 factor is not meaningful, just omega_t
## Omega_h for 1 factor is not meaningful, just omega_t
## Omega_h for 1 factor is not meaningful, just omega_t
## Omega_h for 1 factor is not meaningful, just omega_t

```

##	chisq.scaled	df.scaled	pvalue.scaled	cfi.scaled	tli.scaled
##	75.977	24.000	0.000	0.898	0.847
##	rmsea.scaled	rmsea.ci.lower.scaled	rmsea.ci.upper.scaled	rmsea.pvalue.scaled	srmr
##	0.092	0.069	0.116	0.002	0.064

9.3 SDR-5 (SDR1-SDR5)

```
sdr_cols <- paste0("SDR", 1:5)
has_sdr <- all(sdr_cols %in% names(dat))

if (!has_sdr) {
  cat("\n**SDR-5 section skipped:** SDR1-SDR5 not found.\n")
} else {
  sdr_df <- dat[keep, sdr_cols] %>% as_tibble()
  a_sdr <- psych::alpha(sdr_df)
  ap_sdr <- alpha_poly_likert(sdr_df)
  wt_sdr <- omega_total_poly_likert(sdr_df)

  print_tbl(tibble(
    scale = "SDR-5",
    k_items = 5,
    alpha_raw = a_sdr$total$raw_alpha,
    alpha_poly = ap_sdr$total$raw_alpha,
    omega_total = wt_sdr
  ), digits = 3, caption = "SDR-5 reliability (alpha, polychoric alpha, omega total).")

  # Optional CFA: 1 factor
  sdr_model <- 'SDR =~ SDR1 + SDR2 + SDR3 + SDR4 + SDR5'
  sdr_fit <- try(lavaan::cfa(sdr_model, data = sdr_df, ordered = colnames(sdr_df), estimator = "WLSMV"), silent = TRUE)
  if (!inherits(sdr_fit, "try-error")) {
    print(fitMeasures(sdr_fit, c(
      "chisq.scaled", "df.scaled", "pvalue.scaled", "cfi.scaled",
      "tli.scaled", "rmsea.scaled", "rmsea.ci.lower.scaled",
      "rmsea.ci.upper.scaled", "rmsea.pvalue.scaled", "srmr"
    )))
  } else {
    cat("\nSDR-5 CFA skipped (model failed to converge).\n")
  }
}
```

```
## Some items ( SDR1 SDR5 ) were negatively correlated with the first principal component and
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' option

## Some items ( SDR1 SDR5 ) were negatively correlated with the first principal component and
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' option

## Omega_h for 1 factor is not meaningful, just omega_t
```

##	chisq.scaled	df.scaled	pvalue.scaled	cfi.scaled	tli.scaled
##	61.401	5.000	0.000	0.640	0.279
##	rmsea.scaled	rmsea.ci.lower.scaled	rmsea.ci.upper.scaled	rmsea.pvalue.scaled	srmr
##	0.206	0.162	0.254	0.000	0.103

10 Step 9 – Cross-construct correlations (HPT, KN, FR-LF, KSA-3, SDR-5)

```
# Build scale scores that exist in your data (gracefully skipping any missing block)
scales_list <- list(
  HPT_CTX6 = hpt_scores$HPT_CTX6,
  HPT_TOT9 = hpt_scores$HPT_TOT9,
  HPT_POP  = hpt_scores$POP_raw, # presentism foil (higher = worse)
  HPT_ROA  = hpt_scores$ROA,
  HPT_CONT = hpt_scores$CONT
)

# optional blocks
kn_cols <- paste0("KN", 1:6); has_kn <- all(kn_cols %in% names(dat))
fr_cols <- c(paste0("RD", 1:3), paste0("NS", 1:3)); has_fr <- all(fr_cols %in% names(dat))
ksa_cols <- c(paste0("A", 1:3), paste0("U", 1:3), paste0("K", 1:3)); has_ksa <- all(ksa_cols %in% names(dat))
sdr_cols <- paste0("SDR", 1:5); has_sdr <- all(sdr_cols %in% names(dat))

if (has_kn) {
  scales_list$KN_total <- rowSums(dat[keep, kn_cols], na.rm = TRUE)
}

if (has_fr) {
  fr_df <- dat[keep, fr_cols]
  scales_list$FR_RD <- scale_mean(fr_df, paste0("RD", 1:3), min_answered = 2)
  scales_list$FR_NS <- scale_mean(fr_df, paste0("NS", 1:3), min_answered = 2)
  scales_list$FR_total <- scale_mean(fr_df, fr_cols, min_answered = 4)
}

if (has_ksa) {
  ksa_df <- dat[keep, ksa_cols]
  scales_list$KSA_A <- scale_mean(ksa_df, paste0("A", 1:3), min_answered = 2)
  scales_list$KSA_U <- scale_mean(ksa_df, paste0("U", 1:3), min_answered = 2)
  scales_list$KSA_K <- scale_mean(ksa_df, paste0("K", 1:3), min_answered = 2)
  scales_list$KSA_total <- scale_mean(ksa_df, ksa_cols, min_answered = 7)
}
```

```

if (has_sdr) {
  sdr_df <- dat[keep, sdr_cols]
  scales_list$SDR_total <- scale_mean(sdr_df, sdr_cols, min_answered = 4)
}

scales_df <- as_tibble(scales_list)

# Pairwise complete correlations
cors <- cor(scales_df, use = "pairwise.complete.obs")

print_tbl(round(cors, 3), caption = "Cross-construct correlations (pairwise complete).")

```

Table 4: Cross-construct correlations (pairwise complete).

	HPT_CTX6	HPT_TOT9	HPT_POP	HPT_ROA	HPT_CONT	KN_total	FR_RD	FR_NS	FR_total	KSA_A	KSA_U	KSA_K
HPT_CTX6	1.000	0.901	-0.770	0.335	0.826	0.340	-0.062	0.003	-0.037	0.008	-0.072	-0.011
HPT_TOT9	0.901	1.000	-0.645	0.711	0.787	0.378	-0.036	-0.003	-0.024	0.038	-0.040	0.026
HPT_POP	-0.770	-0.645	1.000	-0.153	-0.276	-0.307	0.091	0.063	0.093	0.065	0.141	0.115
HPT_ROA	0.335	0.711	-0.153	1.000	0.369	0.268	0.022	-0.010	0.007	0.070	0.030	0.074
HPT_CONT	0.826	0.787	-0.276	0.369	1.000	0.241	-0.012	0.060	0.027	0.070	0.017	0.086
KN_total	0.340	0.378	-0.307	0.268	0.241	1.000	-0.062	-0.106	-0.095	0.037	0.012	0.082
FR_RD	-0.062	-0.036	0.091	0.022	-0.012	-0.062	1.000	0.423	0.839	0.379	0.415	0.359
FR_NS	0.003	-0.003	0.063	-0.010	0.060	-0.106	0.423	1.000	0.847	0.353	0.371	0.228
FR_total	-0.037	-0.024	0.093	0.007	0.027	-0.095	0.839	0.847	1.000	0.433	0.467	0.348
KSA_A	0.008	0.038	0.065	0.070	0.070	0.037	0.379	0.353	0.433	1.000	0.339	0.423
KSA_U	-0.072	-0.040	0.141	0.030	0.017	0.012	0.415	0.371	0.467	0.339	1.000	0.387
KSA_K	-0.011	0.026	0.115	0.074	0.086	0.082	0.359	0.228	0.348	0.423	0.387	1.000
KSA_total	-0.031	0.012	0.139	0.077	0.076	0.056	0.499	0.414	0.540	0.790	0.723	0.787
SDR_total	-0.027	-0.017	0.041	0.007	-0.004	0.004	-0.068	-0.242	-0.185	-0.149	-0.174	-0.012

11 Reproducibility appendix

```
sessionInfo()
```

```

## R version 4.6.1 (2026-06-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.5 LTS
##
## Matrix products: default

```

```

## BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0 LAPACK version 3.12.0
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C              LC_TIME=cs_CZ.UTF-8      LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=cs_CZ.UTF-8   LC_MESSAGES=en_US.UTF-8   LC_PAPER=cs_CZ.UTF-8     LC_NAME=C
## [9] LC_ADDRESS=C             LC_TELEPHONE=C            LC_MEASUREMENT=cs_CZ.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Prague
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] kableExtra_1.4.1 knitr_1.51      semTools_0.5-9  lavaan_0.7-2    psych_2.6.5     lubridate_1.9.5
## [7] forcats_1.0.1    stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2     readr_2.2.0     tidyr_1.3.2
## [13] tibble_3.3.1     ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] GPARotation_2026.8-2 generics_0.1.4    xml2_1.6.0        stringi_1.8.7     lattice_0.23-1
## [6] hms_1.1.4         digest_0.6.39     magrittr_2.0.5     evaluate_1.0.5     grid_4.6.1
## [11] timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0      tinytex_0.61       viridisLite_0.4.3
## [16] scales_1.4.0       pbivnorm_0.6.0    textshaping_1.0.5 mnormt_2.1.2       cli_3.6.6
## [21] rlang_1.3.0        withr_3.0.3       yaml_2.3.12        otel_0.2.0         tools_4.6.1
## [26] parallel_4.6.1     tzdb_0.5.0        vctrs_0.7.3        R6_2.6.1           stats4_4.6.1
## [31] lifecycle_1.0.5    MASS_7.3-66       pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6
## [36] glue_1.8.1         systemfonts_1.3.2 xfun_0.60          tidyselect_1.2.1   rstudioapi_0.19.0
## [41] farver_2.1.2       htmltools_0.5.9   nlme_3.1-171       labeling_0.4.3     svglite_2.2.2
## [46] rmarkdown_2.32     compiler_4.6.1    S7_0.2.2           quadprog_1.5-8

```